OPTIONAL EXERCISES

Exercise 4: Log-linearization

Log-linearize the following equations.

1. $\frac{W_t}{P_t} = \alpha \frac{A_t}{N_t^{1-\alpha}}$
2. $Y_t = e^{Z_t} K_{t-1}^{\theta} N_t^{1-\theta}$
3. $C_t + K_t = Y_t + (1-\delta) K_{t-1}$
4. $Y_t = A_t [\alpha K_{t-1}^{\rho} + (1-\alpha) N_t^{\rho}]^{\frac{1}{\rho}}$
5. $\ln(Z_t) = Z_0 + \rho \ln(Z_{t-1}) + \epsilon_t$
6. $\frac{1}{C_t} = \beta \mathbb{E}_t [\frac{R_{t+1}}{C_{t+1}}]$, where $\mathbb{E}_t$ is the expectation operator

Exercise 5: Consumption Dynamics

Suppose that a consumer solves the following problem:

$$\max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \, u(c_t)$$

$$\text{s.t.} \begin{cases} c_t + a_{t+1} = (1+r)\, a_t + y_t \\ c_t \geq 0 \\ a_0 \ \text{ given} \end{cases}$$

where a_t denotes the amount of 1-period bonds held at the beginning of period t, and yielding a constant rate of return r. y_t is labor income in period t (assumed to be exogenous and known with perfect foresight), u represents the period utility function (discounted intertemporally according to β) assumed to be differentiable, strictly increasing and strictly concave. We also assume that $\lim_{c \to 0} u'(c) = \infty$, $\lim_{c \to \infty} u'(c) = 0$, and that the following so-called no-Ponzi games (NPG hereafter) condition holds:

$$\lim_{T \to \infty} \left(\frac{1}{1+r} \right)^T a_T = 0$$

1. Formulate the consumer problem in recursive form: identify the state variable, the control variables, and write down the Bellman equation.
2. Find the first-order necessary conditions of the consumer problem. Then, exploit the envelope condition (one period ahead) to express the first order conditions in terms of the variables of the model, by eliminating the multiplier. Provide the economic intuition for the resulting Euler equation.

[HINT: you don't need to substitute the constraint].

3. Starting from the flow budget constraint obtain the lifetime budget constraint, expressing the present discounted value of lifetime consumption as a function of the present discounted value of lifetime resources.

   [HINT: the flow budget constraint at $t = 0$ and $t = 1$ is:

$$a_1 = (1 + r)\, a_0 + y_0 - c_0$$
$$a_2 = (1 + r)\, a_1 + y_1 - c_1$$

   Combine the previous equations to get rid of a_1. Iterate on this procedure until $t = T$, take the limit $T \to \infty$, and use the NPG condition].

4. Assume that $\beta(1 + r) = 1$ and $a_0 = 0$. How is the optimal consumption path in this case? Use the lifetime budget constraint to express per-period consumption as a function of lifetime resources. Finally, assume also $y_t = \bar{y}$ for every $t \geq 0$, and derive the the implied optimal asset sequence. Provide an economic intuition for your results.

   [HINT: you may need the following result: since $\beta < 1$, then $\sum_{j=0}^{\infty} \beta^j = \frac{1}{1-\beta}$].

5. To understand the role of the NPG, consider this alternative path for consumption $\{\hat{c}_t\}_{t\geq 0}$ and asset holdings $\{\hat{a}_t\}_{t\geq 0}$: suppose $\hat{a}_0 = 0$, and the household borrows an (arbitrarily small) amount $\hat{a}_1 = -\varepsilon < 0$ at $t = 0$, so that

$$\hat{c}_0 = \bar{y} + \varepsilon$$

   while $\hat{c}_t = \bar{y}$ for every other period ($t \geq 1$). Hence, it must be the case that:

$$\hat{a}_1 = -\varepsilon$$
$$\hat{a}_2 = -(1 + r)\, \varepsilon$$
$$\ldots$$
$$\hat{a}_T = -(1 + r)^{T-1}\, \varepsilon$$

   Is the path $c_t = \bar{y}$ for all $t \geq 0$ still optimal? Is the NPG violated? Briefly discuss the underlying economic intuition.

Solution

1. The (endogenous) state variable of the problem is a_t. The control variables are c_t and k_{t+1}. The Bellman equation is

$$V(a_t) = \max_{\{c_t, a_{t+1}\}} \left[u(c_t) + \beta\, V(a_{t+1})\right]$$

$$\text{s.t. } c_t + a_{t+1} = (1+r)\, a_t + y_t$$

2. The first order conditions (FOC) are:

$$\left[\frac{\partial}{\partial\, c_t}\right] : \quad u'(c_t) = \lambda_t$$
$$\left[\frac{\partial}{\partial\, a_{t+1}}\right] : \quad \beta\, \frac{\partial\, V(a_{t+1})}{\partial\, a_{t+1}} = \lambda_t$$

where λ_t is the multiplier associated with the period-t budget constraint.

The envelope condition gives:

$$\frac{\partial\, V(a_t)}{\partial\, a_t} = \lambda_t\, (1+r)$$

Leading the envelope condition one period ahead, and combining it with the FOC to get rid of the multiplier λ_t, yields the following Euler equation:

$$u'(c_t) = \beta(1+r)u'(c_{t+1})$$

At the optimum, the consumer balances the marginal costs of reducing consumption in period t by 1 unit (measured by the marginal utility of consumption) with the marginal benefits of raising it in period $t+1$ by $(1+r)$ units. Moreover, since the additional consumption would take place in the following period, its marginal benefits are discounted by β.

3. Proceeding by forward iteration gives the following period-T budget constraint

$$\left(\frac{1}{1+r}\right)^T a_T = a_0 + \left(\frac{1}{1+r}\right) \sum_{t=0}^{T} \left(\frac{1}{1+r}\right)^t (y_t - c_t).$$

Letting $T \to \infty$, imposing the NPG condition $\lim_{T\to\infty} \left(\frac{1}{1+r}\right)^T a_T = 0$, and rearranging, yields the following lifetime budget constraint

$$\left(\frac{1}{1+r}\right) \sum_{t=0}^{\infty} \left(\frac{1}{1+r}\right)^t c_t = a_0 + \left(\frac{1}{1+r}\right) \sum_{t=0}^{\infty} \left(\frac{1}{1+r}\right)^t y_t$$

which states that the present discounted value of lifetime consumption should be equal to the present discounted value of available resources.

As such, the NPG is an additional constraint of the problem, to be imposed ex-ante. Even if it resembles the transversality condition (TVC), this latter is instead an optimality condition to be imposed only on the optimal amount of assets a^*.

4. From the Euler equation we get, for the case $\beta(1+r) = 1$, that $u'(c_t) = u'(c_{t+1})$ at every period $t \geq 0$. As u is assumed to be strictly increasing this implies that, at

the optimum, consumption exhibits a constant path with $c_{t+1} = c_t = \bar{c}$. Therefore, from the lifetime budget constraint obtained in (2), we get:

$$\bar{c} = \frac{r}{1+r}\left[\sum_{t=0}^{\infty}\left(\frac{1}{1+r}\right)^t y_t\right].$$

By assuming a constant level of income $\bar{y}$ in every period t, we obtain:

$$\bar{c} = \bar{y}\frac{r}{1+r}\frac{1+r}{r} = \bar{y}.$$

Therefore, the optimal asset sequence is $a_t^* = 0$ at every period $t \geq 1$. This is due to the fact that when $\beta = \frac{1}{1+r}$ and $y_t = \bar{y}$ for all t, there are no advantages from increasing consumption (by borrowing) in any period, as the benefits of increasing consumption are completely offset by the loss of utility derived from repaying the loan in the following periods. Then, at the optimum, the household consumes a constant fraction of his lifetime resources $\bar{c} = \bar{y}$.

5. $c_t = \bar{y}$ for all $t \geq 0$ would not be optimal any more, as the household can be better off by increasing consumption by an arbitrarily small amount (only) at $t = 0$ and rolling over the incurred debt forever. Notice, though, that this violates the NPG condition, which rules out such rolling-over behaviors. Indeed,

$$\hat{a}_T = -(1+r)^{T-1}\varepsilon$$

is equivalent to

$$\left(\frac{1}{1+r}\right)^{T-1}\hat{a}_T = -\varepsilon$$

and, as $T \to \infty$, we have

$$\lim_{T\to\infty}\left(\frac{1}{1+r}\right)^{T}\hat{a}_T = -\varepsilon < 0$$

a violation of the NPG condition. By ruling out unbalanced borrowing, the NPG condition ensures that the net present value of the total stream of consumption is equal to the present value of the available resources.

Notice that this condition should hold for all feasible sequences asset holdings (not only for the optimal one). Otherwise, the maximization problem would not have a solution, as any consumption path (and, thus, the sequence of period utilities) could be improved by increasing the amount of borrowing in each period.